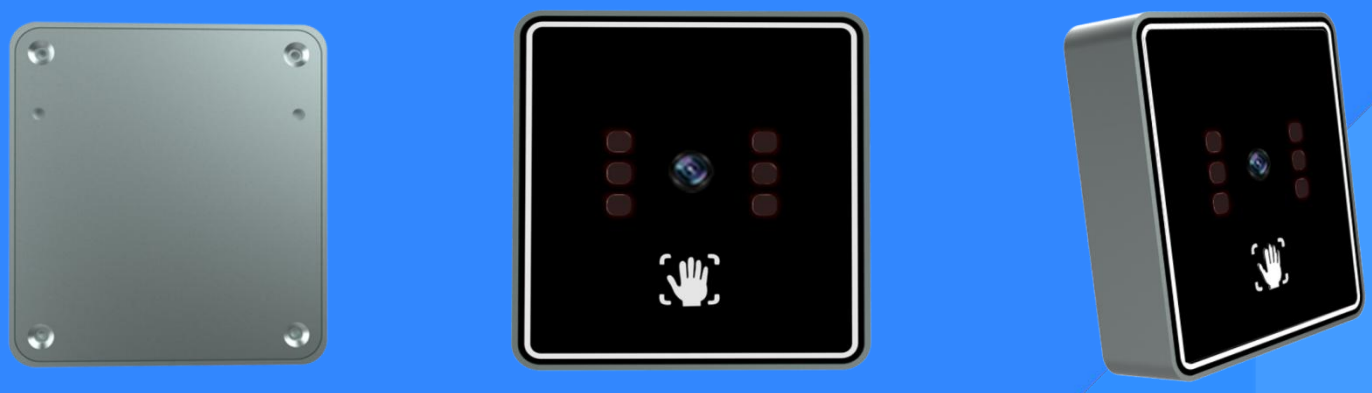
Monocular Palm Vein Capture and Recognition Module



TM-P

【Monocular Palm Vein Capture and Recognition Module】

■ Product Description

TM-P is the new multi-angle anti-exposure palm vein recognition terminal of Tendcent.The product adopts a high performance lens FEL:3.2mm F/NO:2.4 to deal with the impact of backlight environment on the speed of palm recognition.Scanning speed 0.1s can quickly detect the palm of the hand in a short period of time.The product supports a variety of systems Windows/Linux/Android, can be used with rk3568, rk3288 and other Android motherboards, provide a complete set of dome and development files to make independent development time is greatly reduced.Palm vein module has a wide range of applications, it can be used in the access control system of community, industrial park, office building, as well as library lending and returning machines, access gates of underground and bus, security check channel of airport and railway station, etc. It can also be applied in the canteen consumption payment, lift control and other scenes, which provides reliable technical support for personnel management, security control and convenient payment.

In terms of security performance, palm vein recognition, as a highly secure biometric technology, is unique and difficult to replicate.The TM-P module takes advantage of the fact that each person's palm vein pattern is unique and located inside the skin, making it difficult to be stolen or forged, providing customers with reliable security against unauthorised access and intrusion.At the same time, the module is equipped with a live body detection function, which can effectively defend against fake attacks such as pictures and mobile phone screens.

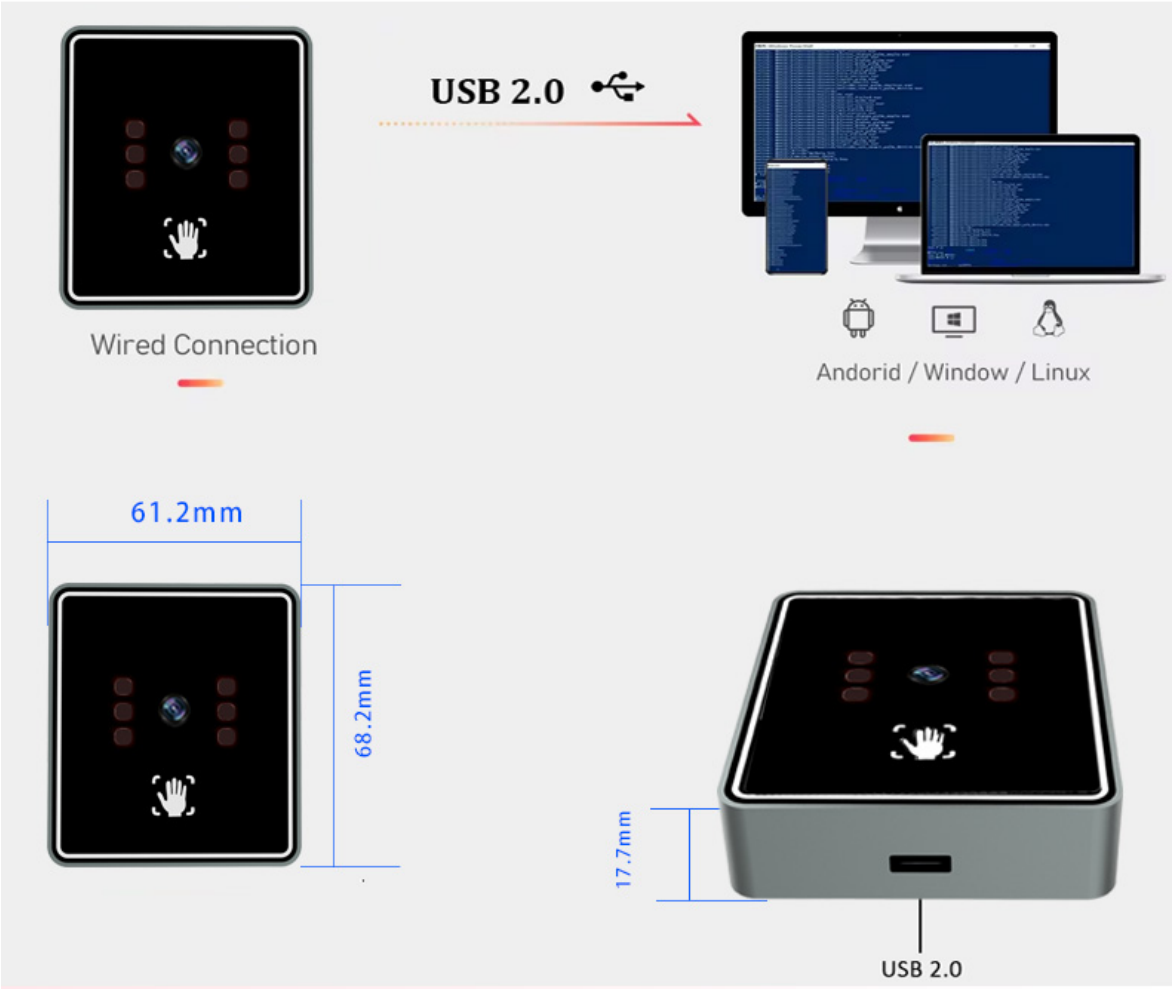
Product Advantages

- Non contact, module able to recognize palm vein at a distance of 5cm-20cm;
- Adopt embedded palm vein chip, support a variety of development overall;
- Size: 38*38*10.5mm / 32*32*10.5mm;
- Windows, Android, Linux
- Highly accurate recognition, 99.5% certainty in monocular liveness detection;
- Support palm vein recognition、Multi-angle acquisition recognition method;

Basic Parameters

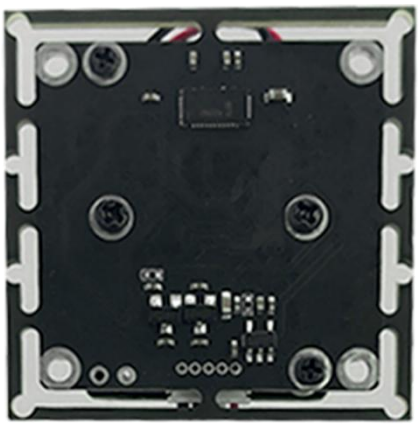
Model No.	TM-P
Size	38*38*10.5mm / 32*32*10.5
Resolution	1920 H x 1080 V
Pixel size	2.8*2.8 μm
Dynamic Range	NC
Output format	MJPEG, YUV2
Field of view	DFOV: 92°
Lens	FEL:3.2mm F/NO:2.4
Scanning speed	0.1S
Focus type	FF/ MF
Shutter type	Electronic Rolling shutter
Palm vein capacity	10000
Interface	5 PIN 1.25MM Socket
Minimum illumination	0.01Lux@(F1.2,AGC ON)
Support system	Windows, Android, Linux, Wince, Mac OS
Operating voltage	USB DC +5V
Operating current	<200mA
Operating temperature	-20℃ ~ 60℃
Storage temperature	-20℃ ~ 85℃
Cable length	TBD

Support connection to various system devices



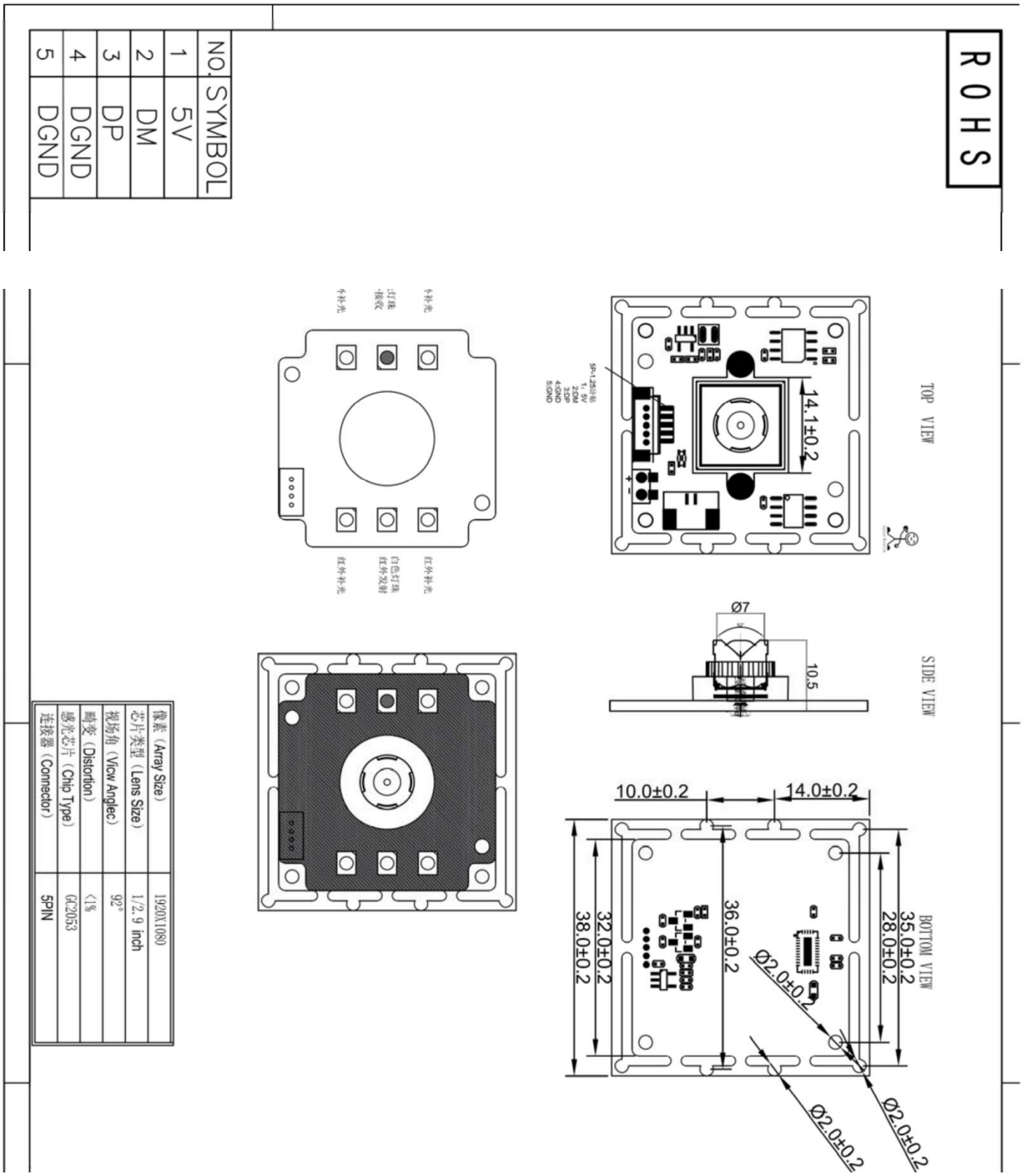
Palm vein recognition	
Supported system	Windows/Linux/Android
Angle of use	360° recognition
Palm vein capacity	10000
Reject rate	<0.01%
Recognition rate	<0.00001%
Authentication speed	1 :N< 1s
Comparison mode	1:1 1: N
Recognition distance	5-20cm
Light processing	anti-exposure

形状和尺寸 Shape and Dimensions (Unit: mm)



PIN 脚 Pin No.	信号 Signal Name	描述 Description
1	VCC	5V Power
2	DM	USB2.0 Data+
3	DP	USB2.0 Data-
4	GND	GND
5	GND	GND

PIN 脚定义 PIN Definition



NO.	SYMBOL
1	5V
2	DM
3	DP
4	DGND
5	DGND

像素（Array Size）	1920X1080
芯片类型（Lens Size）	1/2.9 inch
视场角（View Anglec）	92°
畸变（Distortion）	<1%
感光芯片（Chip Type）	GC2053
连接器（Connector）	5PIN

Business premises

Gates